

Table of Contents

Dimension	3
MedID_Dimension	3
LabReference_Dimension	5
SampleID_Dimension	6
Collected_Dimension	7
Time_Dimension.....	8
Reported_Time_Dimension	9
Fact.....	10
Fact_Urine.....	10
Fact_CBC	12
Fact_Platelet_Profile	14
Fact_Lipid_Profile	16
Fact_Liver_Function.....	18
Fact_Kidney_Function.....	20
Fact_Iron_Profile	22
Fact_HbA1c.....	24
Fact_Urine_ACR.....	26
Fact_Calcium_Phos	28
Fact_Thyroid_Profile	30
Fact_Glucose_Fasting.....	32
Fact_Glucose_PP.....	34
Fact_Glucose_Diagnopath	36
View	38
View_Fact_Urine.....	38
View_Fact_CBC	39
View_Fact_Platet_Profile	40
View_Fact_Lipid_Profile	41
View_Fact_Liver_Function.....	42
View_Fact_Kidney_Function.....	43
View_Fact_Iron_Profile	44

View_Fact_HbA1C	45
View_Fact_Urine_ACR.....	46
View_Fact_Calcium_Phos	47
View_Fact_Thyroid_Profile	48
View_Fact_Glucose_Fasting.....	49
View_Fact_Glucose_PP.....	50
View_Fact_Glucose_Diagnopath	51
Schema	52
MedID_Dimension	52
LabReference_Dimension	52
SampleID_Dimension	52
Collected_Dimension	52
Time_Dimension.....	52
Reported_Time_Dimension	53
Fact_Urine.....	53
Fact_CBC	53
Fact_Platelet_Profile	54
Fact_Lipid_Profile	54
Fact_Liver_Function.....	55
Fact_Kidney_Function.....	55
Fact_Iron_Profile	56
Fact_HbA1c.....	56
Fact_Urine_ACR.....	56
Fact_Calcium_Phos	56
Fact_Thyroid_Profile	57
Fact_Glucose_Fasting.....	57
Fact_Glucose_PP.....	57
Fact_Glucose_Diagnopath	57

Dimension

MedID_Dimension

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.MedID_Dimension`
(
  ID INT64 NOT NULL,
  Original_MedID STRING NOT NULL,
  LabReference STRING NOT NULL,
  First_Name STRING NOT NULL,
  Last_Name STRING NOT NULL,
  Age INT64 NOT NULL,
  Age_Category STRING NOT NULL,
  Gender STRING NOT NULL,
  Diet STRING NOT NULL,
  Registration_Weight FLOAT64 NOT NULL
)
AS
WITH Age_Ranges AS (
  -- Parse the text range '25-29' or '25–29' into numeric Start and End ages
  SELECT
    Category,
    -- Handling both en-dash and hyphen
    CAST(SPLIT(REPLACE(`Age_Range`, '-', '-'), '-') [OFFSET(0)] AS INT64) AS Start_Age,
    CAST(SPLIT(REPLACE(`Age_Range`, '-', '-'), '-') [OFFSET(1)] AS INT64) AS End_Age
  FROM
    `bigquery-tutorial-480009.A2.AgeCategories`
)
SELECT
  -- Generate sequential ID (Surrogate Key)
  ROW_NUMBER() OVER (ORDER BY m.MedID) AS ID,

  -- Original ID cast to String
  CAST(m.MedID AS STRING) AS Original_MedID,

  -- Use COALESCE to ensure no null values are inserted
  COALESCE(m.LabReference, 'Unknown') AS LabReference,
  COALESCE(m.`First Name`, 'Unknown') AS First_Name,
  COALESCE(m.`Last Name`, 'Unknown') AS Last_Name,
  COALESCE(m.Age, -1) AS Age,

  -- Join results from the Age_Ranges CTE
  COALESCE(a.Category, 'Unknown') AS Age_Category,

  -- Map Zender to Gender and handle nulls
  COALESCE(m.Gender, 'Unknown') AS Gender,
```

```
COALESCE(m.Diet, 'Unknown') AS Diet,  
COALESCE(m.`Registration Weight`, 0.0) AS Registration_Weight  
  
FROM  
  `bigquery-tutorial-480009.A2.MedIDDetails` m  
LEFT JOIN  
  Age_Ranges a  
ON  
  m.Age BETWEEN a.Start_Age AND a.End_Age;
```

LabReference_Dimension

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.LabReference_Dimension`
(
  ID INT64 NOT NULL,
  LabReference STRING NOT NULL
)
AS
WITH All_Unique_Refs AS (
  -- 1. Get unique LabReferences from the Medical Records
  SELECT DISTINCT LabReference
  FROM `bigquery-tutorial-480009.A2.Medical_Records`
  WHERE LabReference IS NOT NULL

  UNION DISTINCT

  -- 2. Get unique LabReferences from the Patient Details
  SELECT DISTINCT LabReference
  FROM `bigquery-tutorial-480009.A2.MedIDDetails`
  WHERE LabReference IS NOT NULL
)

SELECT
  -- Generate a sequential ID (Surrogate Key)
  ROW_NUMBER() OVER (ORDER BY LabReference) AS ID,

  -- The Lab Reference Code
  LabReference
FROM
  All_Unique_Refs;
```

SampleID_Dimension

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.SampleID_Dimension`
(
  ID INT64 NOT NULL,
  Sample_ID STRING NOT NULL
)
AS
SELECT
  -- Generates a sequential ID (1, 2, 3...)
  ROW_NUMBER() OVER (ORDER BY `Sample ID`) AS ID,

  -- Cast the original integer ID to String
  CAST(`Sample ID` AS STRING) AS Sample_ID
FROM (
  -- Select distinct samples to ensure uniqueness
  SELECT DISTINCT `Sample ID`
  FROM `bigquery-tutorial-480009.A2.Medical_Records_Cleaned`
  WHERE `Sample ID` IS NOT NULL
);
```

Collected_Dimension

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Collected_Dimension`
(
  ID INT64 NOT NULL,
  Date DATE NOT NULL
)
AS
SELECT
  -- Auto-incremental ID (1, 2, 3...)
  ROW_NUMBER() OVER (ORDER BY Collected) AS ID,

  -- The actual Date value
  Collected AS Date
FROM (
  -- Get unique dates from the cleaned fact table
  SELECT DISTINCT Collected
  FROM `bigquery-tutorial-480009.A2.Medical_Records_Cleaned`
  WHERE Collected IS NOT NULL
);
```

Time_Dimension

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Time_Dimension`
(
  ID INT64 NOT NULL,
  Time TIME NOT NULL
)
AS
SELECT
  -- Auto-incremental ID (1, 2, 3...)
  ROW_NUMBER() OVER (ORDER BY Time) AS ID,

  -- The actual Time value (24-hour format)
  Time
FROM (
  -- Get unique Collection Times from the cleaned fact table
  SELECT DISTINCT Time
  FROM `bigquery-tutorial-480009.A2.Medical_Records_Cleaned`
  WHERE Time IS NOT NULL
);
```


Reported_Time_Dimension

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Reported_Time_Dimension`
(
  ID INT64 NOT NULL,
  Reported_Time TIME NOT NULL
)
AS
SELECT
  -- Auto-incremental ID (1, 2, 3...)
  ROW_NUMBER() OVER (ORDER BY `Reported Time`) AS ID,

  -- The actual Reported Time value
  `Reported Time` AS Reported_Time
FROM (
  -- Get unique Reported Times from the cleaned fact table
  SELECT DISTINCT `Reported Time`
  FROM `bigquery-tutorial-480009.A2.Medical_Records_Cleaned`
  WHERE `Reported Time` IS NOT NULL
);
```

Fact

Fact_Urine

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Urine`
(
  -- Foreign Keys (Linking to Dimensions)
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Urine Facts
  Urine_Colour STRING NOT NULL,
  Appearance STRING NOT NULL,
  Specific_Gravity FLOAT64 NOT NULL,
  pH FLOAT64 NOT NULL,
  Proteins STRING NOT NULL,
  Glucose STRING NOT NULL,
  Bilirubin STRING NOT NULL,
  Ketones STRING NOT NULL,
  Blood STRING NOT NULL,
  Urobilinogen STRING NOT NULL,
  Nitrites STRING NOT NULL,
  WBC_Pus_Cells STRING NOT NULL,
  RBC STRING NOT NULL,
  Epithelial_Cells STRING NOT NULL,
  Casts STRING NOT NULL,
  Crystals STRING NOT NULL,
  Others STRING NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Use COALESCE to handle any missing links, default to -1)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- URINE FACTS (Handle Nulls with Defaults)
  COALESCE(f.`Urine Colour`, 'Unknown') AS Urine_Colour,
  COALESCE(f.Appearance, 'Unknown') AS Appearance,
  COALESCE(f.`Specific Gravity`, 0.0) AS Specific_Gravity,
```

```

COALESCE(f.pH, 0.0) AS pH,
COALESCE(f.Proteins, 'Unknown') AS Proteins,
COALESCE(f.Glucose, 'Unknown') AS Glucose,
COALESCE(f.Bilirubin, 'Unknown') AS Bilirubin,
COALESCE(f.Ketones, 'Unknown') AS Ketones,
COALESCE(f.Blood, 'Unknown') AS Blood,
COALESCE(f.Urobilinogen, 'Unknown') AS Urobilinogen,
COALESCE(f.Nitrites, 'Unknown') AS Nitrites,

-- Map the exact column names from your schema to clean names
COALESCE(f.`WBC _Pus Cells_ __HPF_`, 'Unknown') AS WBC_Pus_Cells,
COALESCE(f.RBC, 'Unknown') AS RBC,
COALESCE(f.`Epithelial Cells __HPF_`, 'Unknown') AS Epithelial_Cells,

COALESCE(f.Casts, 'Unknown') AS Casts,
COALESCE(f.Crystals, 'Unknown') AS Crystals,
COALESCE(f.Others, 'Unknown') AS Others

FROM
`bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
-- Note: Casting Fact IDs to STRING to match Dimension definitions
LEFT JOIN
`bigquery-tutorial-480009.A2.MedID_Dimension` d_med
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID

LEFT JOIN
`bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab
  ON f.LabReference = d_lab.LabReference

LEFT JOIN
`bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID

LEFT JOIN
`bigquery-tutorial-480009.A2.CollectedException` d_coll
  ON f.CollectedException = d_coll.Date

LEFT JOIN
`bigquery-tutorial-480009.A2.Time_Dimension` d_time
  ON f.Time = d_time.Time

LEFT JOIN
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep
  ON f.`Reported Time` = d_rep.Reported_Time;

```

Fact_CBC

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_CBC`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- CBC Facts (Measures)
  Hemoglobin FLOAT64 NOT NULL,
  RBC_Count FLOAT64 NOT NULL,
  Hematocrit FLOAT64 NOT NULL,
  MCV FLOAT64 NOT NULL,
  MCH INT64 NOT NULL,
  MCHC FLOAT64 NOT NULL,
  RDW_CV FLOAT64 NOT NULL,
  RDW_SD FLOAT64 NOT NULL,
  WBC_Count INT64 NOT NULL,
  Neutrophils INT64 NOT NULL,
  Lymphocytes INT64 NOT NULL,
  Eosinophils INT64 NOT NULL,
  Monocytes INT64 NOT NULL,
  Basophils FLOAT64 NOT NULL,      -- Note: Schema defines this as FLOAT64
  Abs_Neutrophils FLOAT64 NOT NULL,
  Abs_Lymphocytes FLOAT64 NOT NULL,
  Abs_Monocytes FLOAT64 NOT NULL,
  Abs_Eosinophils FLOAT64 NOT NULL,
  Abs_Basophils FLOAT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- CBC FACTS (Measures)
  COALESCE(f.`Hemoglobin _g_dL`, 0.0) AS Hemoglobin,
  COALESCE(f.`RBC Count _mil_uL`, 0.0) AS RBC_Count,
  COALESCE(f.`Hematocrit %`, 0.0) AS Hematocrit,
  COALESCE(f.`MCV _fL`, 0.0) AS MCV,
```

```

COALESCE(f.`MCH _pg_`, 0) AS MCH,
COALESCE(f.`MCHC _g_dL_`, 0.0) AS MCHC,
COALESCE(f.`RDW-CV %`, 0.0) AS RDW_CV,
COALESCE(f.`RDW-SD _fL_`, 0.0) AS RDW_SD,
COALESCE(f.`WBC _cells_μL_`, 0) AS WBC_Count,
COALESCE(f.`Neutrophils %`, 0) AS Neutrophils,
COALESCE(f.`Lymphocytes %`, 0) AS Lymphocytes,
COALESCE(f.`Eosinophils %`, 0) AS Eosinophils,
COALESCE(f.`Monocytes %`, 0) AS Monocytes,
COALESCE(f.`Basophils %`, 0.0) AS Basophils,
COALESCE(f.`Abs Neutrophils`, 0.0) AS Abs_Neutrophils,
COALESCE(f.`Abs Lymphocytes`, 0.0) AS Abs_Lymphocytes,
COALESCE(f.`Abs Monocytes`, 0.0) AS Abs_Monocytes,
COALESCE(f.`Abs Eosinophils`, 0.0) AS Abs_Eosinophils,
COALESCE(f.`Abs Basophils`, 0.0) AS Abs_Basophils

FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
LEFT JOIN
  `bigquery-tutorial-480009.A2.MedID_Dimension` d_med
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID

LEFT JOIN
  `bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab
  ON f.LabReference = d_lab.LabReference

LEFT JOIN
  `bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID

LEFT JOIN
  `bigquery-tutorial-480009.A2.CollectedException` d_coll
  ON f.CollectedException = d_coll.Date

LEFT JOIN
  `bigquery-tutorial-480009.A2.Time_Dimension` d_time
  ON f.Time = d_time.Time

LEFT JOIN
  `bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep
  ON f.`Reported Time` = d_rep.Reported_Time;

```

Fact_Platelet_Profile

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Platelet_Profile`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Platelet Facts (Measures)
  Platelet_Count INT64 NOT NULL,
  MPV FLOAT64 NOT NULL,
  Platelet_RDW INT64 NOT NULL,
  PCT FLOAT64 NOT NULL,
  P_LCR FLOAT64 NOT NULL,
  IMG FLOAT64 NOT NULL,
  IMM FLOAT64 NOT NULL,
  IML FLOAT64 NOT NULL,
  LIC FLOAT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- PLATELET FACTS (Measures)
  COALESCE(f.`Platelet Count __10_3_µL_`, 0) AS Platelet_Count,
  COALESCE(f.`MPV _fL_`, 0.0) AS MPV,
  COALESCE(f.`Platelet RDW %`, 0) AS Platelet_RDW,
  COALESCE(f.`PCT %`, 0.0) AS PCT,
  COALESCE(f.`P-LCR %`, 0.0) AS P_LCR,
  COALESCE(f.`IMG %`, 0.0) AS IMG,
  COALESCE(f.`IMM %`, 0.0) AS IMM,
  COALESCE(f.`IML %`, 0.0) AS IML,
  COALESCE(f.`LIC %`, 0.0) AS LIC

FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.MedID_Dimension` d_med  
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab  
  ON f.LabReference = d_lab.LabReference
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp  
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.CollectedException` d_coll  
  ON f.CollectedException = d_coll.Date
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Time_Dimension` d_time  
  ON f.Time = d_time.Time
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep  
  ON f.`Reported Time` = d_rep.Reported_Time;
```

Fact_Lipid_Profile

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Lipid_Profile`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Lipid Profile Facts (Measures)
  Total_Cholesterol INT64 NOT NULL,
  HDL INT64 NOT NULL,
  LDL FLOAT64 NOT NULL,
  VLDL STRING NOT NULL,          -- Kept as STRING to match source schema
  Triglycerides STRING NOT NULL, -- Kept as STRING to match source schema
  Non_HDL INT64 NOT NULL,
  Total_HDL_Ratio FLOAT64 NOT NULL,
  LDL_HDL_Ratio FLOAT64 NOT NULL,
  HDL_LDL_Ratio FLOAT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- LIPID PROFILE FACTS (Measures)
  -- Renaming to clean column names without units
  COALESCE(f.`Total Cholesterol _mg_dL`, 0) AS Total_Cholesterol,
  COALESCE(f.`HDL _mg_dL`, 0) AS HDL,
  COALESCE(f.`LDL _mg_dL`, 0.0) AS LDL,
  COALESCE(f.`VLDL _mg_dL`, 'Unknown') AS VLDL,
  COALESCE(f.`Triglycerides _mg_dL`, 'Unknown') AS Triglycerides,
  COALESCE(f.`Non-HDL _mg_dL`, 0) AS Non_HDL,
  COALESCE(f.`Total_HDL Ratio`, 0.0) AS Total_HDL_Ratio,
  COALESCE(f.`LDL_HDL Ratio`, 0.0) AS LDL_HDL_Ratio,
  COALESCE(f.`HDL_LDL Ratio`, 0.0) AS HDL_LDL_Ratio
FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f
```



```
-- JOINING DIMENSIONS
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.MedID_Dimension` d_med  
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab  
  ON f.LabReference = d_lab.LabReference
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp  
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.CollectedException` d_coll  
  ON f.CollectedException = d_coll.Date
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.Time_Dimension` d_time  
  ON f.Time = d_time.Time
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep  
  ON f.`Reported Time` = d_rep.Reported_Time;
```

Fact_Liver_Function

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Liver_Function`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Liver Function Facts (Measures)
  Bilirubin_Total FLOAT64 NOT NULL,
  Bilirubin_Direct FLOAT64 NOT NULL,
  Bilirubin_Indirect FLOAT64 NOT NULL,
  ALP INT64 NOT NULL,
  ALT_SGPT INT64 NOT NULL,
  AST_SGOT INT64 NOT NULL,
  GGT INT64 NOT NULL,
  Protein_Total FLOAT64 NOT NULL,
  Albumin FLOAT64 NOT NULL,
  Globulin FLOAT64 NOT NULL,
  A_G_Ratio FLOAT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- LIVER FUNCTION FACTS (Measures)
  COALESCE(f.`Bilirubin Total _mg_dL`, 0.0) AS Bilirubin_Total,
  COALESCE(f.`Bilirubin Direct _mg_dL`, 0.0) AS Bilirubin_Direct,
  COALESCE(f.`Bilirubin Indirect _mg_dL`, 0.0) AS Bilirubin_Indirect,
  COALESCE(f.`ALP _U_L`, 0) AS ALP,
  COALESCE(f.`ALT_SGPT _U_L`, 0) AS ALT_SGPT,
  COALESCE(f.`AST_SGOT _U_L`, 0) AS AST_SGOT,
  COALESCE(f.`GGT _U_L`, 0) AS GGT,
  COALESCE(f.`Protein Total _g_dL`, 0.0) AS Protein_Total,
  COALESCE(f.`Albumin _g_dL`, 0.0) AS Albumin,
  COALESCE(f.`Globulin _g_dL`, 0.0) AS Globulin,

  -- Converting A/G Ratio string to Float for proper analysis
```

```

        COALESCE(SAFE_CAST(f.`A_G Ratio` AS FLOAT64), 0.0) AS A_G_Ratio

FROM
    `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
LEFT JOIN
    `bigquery-tutorial-480009.A2.MedID_Dimension` d_med
    ON CAST(f.MedID AS STRING) = d_med.Original_MedID

LEFT JOIN
    `bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab
    ON f.LabReference = d_lab.LabReference

LEFT JOIN
    `bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp
    ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID

LEFT JOIN
    `bigquery-tutorial-480009.A2.CollectedException` d_coll
    ON f.CollectedException = d_coll.Date

LEFT JOIN
    `bigquery-tutorial-480009.A2.Time_Dimension` d_time
    ON f.Time = d_time.Time

LEFT JOIN
    `bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep
    ON f.`Reported Time` = d_rep.Reported_Time;

```

Fact_Kidney_Function

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Kidney_Function`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Kidney Function Facts (Measures)
  Creatinine FLOAT64 NOT NULL,
  Urea FLOAT64 NOT NULL,
  BUN FLOAT64 NOT NULL,
  BUN_Creatinine_Ratio FLOAT64 NOT NULL,
  Sodium INT64 NOT NULL,
  Potassium FLOAT64 NOT NULL,
  Chloride INT64 NOT NULL,
  Uric_Acid FLOAT64 NOT NULL,
  eGFR FLOAT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- KIDNEY FUNCTION FACTS (Measures)
  COALESCE(f.`Creatinine _mg_dL`, 0.0) AS Creatinine,
  COALESCE(f.`Urea _mg_dL`, 0.0) AS Urea,
  COALESCE(f.`BUN _mg_dL`, 0.0) AS BUN,
  COALESCE(f.`BUN_Creatinine Ratio`, 0.0) AS BUN_Creatinine_Ratio,
  COALESCE(f.`Sodium _mmol_L`, 0) AS Sodium,
  COALESCE(f.`Potassium _mmol_L`, 0.0) AS Potassium,
  COALESCE(f.`Chloride _mmol_L`, 0) AS Chloride,
  COALESCE(f.`Uric Acid _mg_dL`, 0.0) AS Uric_Acid,
  COALESCE(f.`eGFR _mL_min_1_73m²`, 0.0) AS eGFR

FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.MedID_Dimension` d_med  
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab  
  ON f.LabReference = d_lab.LabReference
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp  
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.CollectedException` d_coll  
  ON f.CollectedException = d_coll.Date
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Time_Dimension` d_time  
  ON f.Time = d_time.Time
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep  
  ON f.`Reported Time` = d_rep.Reported_Time;
```

Fact_Iron_Profile

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Iron_Profile`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Iron Profile Facts (Measures)
  Iron INT64 NOT NULL,
  UIBC INT64 NOT NULL,
  TIBC INT64 NOT NULL,
  Transferrin_Saturation FLOAT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- IRON PROFILE FACTS (Measures)
  COALESCE(f.`Iron _µg_dL_`, 0) AS Iron,
  COALESCE(f.`UIBC _µg_dL_`, 0) AS UIBC,
  COALESCE(f.`TIBC _µg_dL_`, 0) AS TIBC,
  COALESCE(f.`Transferrin Saturation %`, 0.0) AS Transferrin_Saturation

FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
LEFT JOIN
  `bigquery-tutorial-480009.A2.MedID_Dimension` d_med
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID

LEFT JOIN
  `bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab
  ON f.LabReference = d_lab.LabReference

LEFT JOIN
  `bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp
```

```
ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.CollectedException` d_coll  
ON f.CollectedException = d_coll.Date
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.Time_Dimension` d_time  
ON f.Time = d_time.Time
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep  
ON f.`Reported Time` = d_rep.Reported_Time;
```

Fact_HbA1c

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_HbA1c`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- HbA1c Facts (Measures)
  HbA1c FLOAT64 NOT NULL,
  Estimated_Avg_Glucose FLOAT64 NOT NULL,
  HbF FLOAT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- HbA1c FACTS (Measures)
  COALESCE(f.`HbA1c` %, 0.0) AS HbA1c,
  COALESCE(f.`Estimated Avg Glucose _mg_dL`, 0.0) AS Estimated_Avg_Glucose,
  COALESCE(f.`HbF` %, 0.0) AS HbF

FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
LEFT JOIN
  `bigquery-tutorial-480009.A2.MedID_Dimension` d_med
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID

LEFT JOIN
  `bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab
  ON f.LabReference = d_lab.LabReference

LEFT JOIN
  `bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID
```


LEFT JOIN

```
`bigquery-tutorial-480009.A2.CollectedException` d_coll  
ON f.CollectedException = d_coll.Date
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Time_Dimension` d_time  
ON f.Time = d_time.Time
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep  
ON f.`Reported Time` = d_rep.Reported_Time;
```

Fact_Urine_ACR

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Urine_ACR`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Urine ACR Facts (Measures)
  Urine_Albumin FLOAT64 NOT NULL,
  Urine_Creatinine FLOAT64 NOT NULL,
  Albumin_Creatinine_Ratio FLOAT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- URINE ACR FACTS (Measures)
  COALESCE(f.`Urine Albumin _mg_L`, 0.0) AS Urine_Albumin,
  COALESCE(f.`Urine Creatinine _mg_dL`, 0.0) AS Urine_Creatinine,
  COALESCE(f.`Albumin_Creatinine Ratio`, 0.0) AS Albumin_Creatinine_Ratio

FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
LEFT JOIN
  `bigquery-tutorial-480009.A2.MedID_Dimension` d_med
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID

LEFT JOIN
  `bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab
  ON f.LabReference = d_lab.LabReference

LEFT JOIN
  `bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.CollectedException` d_coll  
ON f.CollectedException = d_coll.Date
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Time_Dimension` d_time  
ON f.Time = d_time.Time
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep  
ON f.`Reported Time` = d_rep.Reported_Time;
```

Fact_Calcium_Phos

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Calcium_Phos`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Calcium & Phosphorus Facts (Measures)
  Calcium FLOAT64 NOT NULL,
  Phosphorus FLOAT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- CALCIUM & PHOS FACTS (Measures)
  COALESCE(f.`Calcium _mg_dL`, 0.0) AS Calcium,
  COALESCE(f.`Phosphorus _mg_dL`, 0.0) AS Phosphorus

FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
LEFT JOIN
  `bigquery-tutorial-480009.A2.MedID_Dimension` d_med
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID

LEFT JOIN
  `bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab
  ON f.LabReference = d_lab.LabReference

LEFT JOIN
  `bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID

LEFT JOIN
  `bigquery-tutorial-480009.A2.Collected_Dimension` d_coll
```

```
ON f.Collecte d = d_coll.Date
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.Time_Dimension` d_time
```

```
ON f.Time = d_time.Time
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep
```

```
ON f.`Reported Time` = d_rep.Reported_Time;
```

Fact_Thyroid_Profile

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Thyroid_Profile`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Thyroid Profile Facts (Measures)
  TT3 INT64 NOT NULL,
  TT4 FLOAT64 NOT NULL,
  TSH FLOAT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- THYROID PROFILE FACTS (Measures)
  COALESCE(f.`TT3 _ng_dL_`, 0) AS TT3,
  COALESCE(f.`TT4 _µg_dL_`, 0.0) AS TT4,
  COALESCE(f.`TSH _µIU_mL_`, 0.0) AS TSH

FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
LEFT JOIN
  `bigquery-tutorial-480009.A2.MedID_Dimension` d_med
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID

LEFT JOIN
  `bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab
  ON f.LabReference = d_lab.LabReference

LEFT JOIN
  `bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.CollectedException` d_coll  
ON f.CollectedException = d_coll.Date
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Time_Dimension` d_time  
ON f.Time = d_time.Time
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep  
ON f.`Reported Time` = d_rep.Reported_Time;
```

Fact_Glucose_Fasting

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Glucose_Fasting`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Glucose Fasting Facts (Measures)
  Fasting_Glucose INT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- GLUCOSE FASTING FACTS (Measures)
  COALESCE(f.`Fasting Glucose _mg_dL`, 0) AS Fasting_Glucose
FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
LEFT JOIN
  `bigquery-tutorial-480009.A2.MedID_Dimension` d_med
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID

LEFT JOIN
  `bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab
  ON f.LabReference = d_lab.LabReference

LEFT JOIN
  `bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID

LEFT JOIN
  `bigquery-tutorial-480009.A2.Collected_Dimension` d_coll
  ON f.Collected = d_coll.Date
```


LEFT JOIN

```
`bigquery-tutorial-480009.A2.Time_Dimension` d_time  
ON f.Time = d_time.Time
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep  
ON f.`Reported Time` = d_rep.Reported_Time;
```

Fact_Glucose_PP

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Glucose_PP`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Glucose PP Facts (Measures)
  Postprandial_Glucose INT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- GLUCOSE PP FACTS (Measures)
  COALESCE(f.`Postprandial Glucose _mg_dL`, 0) AS Postprandial_Glucose

FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
LEFT JOIN
  `bigquery-tutorial-480009.A2.MedID_Dimension` d_med
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID

LEFT JOIN
  `bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab
  ON f.LabReference = d_lab.LabReference

LEFT JOIN
  `bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID

LEFT JOIN
  `bigquery-tutorial-480009.A2.Collected_Dimension` d_coll
  ON f.Collected = d_coll.Date
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Time_Dimension` d_time  
ON f.Time = d_time.Time
```

LEFT JOIN

```
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep  
ON f.`Reported Time` = d_rep.Reported_Time;
```

Fact_Glucose_Diagnopath

```
CREATE OR REPLACE TABLE `bigquery-tutorial-480009.A2.Fact_Glucose_Diagnopath`
(
  -- Foreign Keys
  MedID_FK INT64 NOT NULL,
  LabReference_FK INT64 NOT NULL,
  SampleID_FK INT64 NOT NULL,
  Collected_FK INT64 NOT NULL,
  Time_FK INT64 NOT NULL,
  Reported_Time_FK INT64 NOT NULL,

  -- Glucose Diagnopath Facts (Measures)
  FBS INT64 NOT NULL,
  PLBS INT64 NOT NULL
)
AS
SELECT
  -- FOREIGN KEYS (Linking to Dimensions)
  COALESCE(d_med.ID, -1) AS MedID_FK,
  COALESCE(d_lab.ID, -1) AS LabReference_FK,
  COALESCE(d_samp.ID, -1) AS SampleID_FK,
  COALESCE(d_coll.ID, -1) AS Collected_FK,
  COALESCE(d_time.ID, -1) AS Time_FK,
  COALESCE(d_rep.ID, -1) AS Reported_Time_FK,

  -- GLUCOSE DIAGNOPATH FACTS (Measures)
  COALESCE(f.`FBS _mg_dL_`, 0) AS FBS,
  COALESCE(f.`PLBS _mg_dL_`, 0) AS PLBS

FROM
  `bigquery-tutorial-480009.A2.Medical_Records_Cleaned` f

-- JOINING DIMENSIONS
LEFT JOIN
  `bigquery-tutorial-480009.A2.MedID_Dimension` d_med
  ON CAST(f.MedID AS STRING) = d_med.Original_MedID

LEFT JOIN
  `bigquery-tutorial-480009.A2.LabReference_Dimension` d_lab
  ON f.LabReference = d_lab.LabReference

LEFT JOIN
  `bigquery-tutorial-480009.A2.SampleID_Dimension` d_samp
  ON CAST(f.`Sample ID` AS STRING) = d_samp.Sample_ID

LEFT JOIN
  `bigquery-tutorial-480009.A2.Collected_Dimension` d_coll
```

```
ON f.Collecte d = d_coll.Date
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.Time_Dimension` d_time
```

```
ON f.Time = d_time.Time
```

```
LEFT JOIN
```

```
`bigquery-tutorial-480009.A2.Reported_Time_Dimension` d_rep
```

```
ON f.`Reported Time` = d_rep.Reported_Time;
```

View

View_Fact_Urine

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Urine` AS
SELECT
  -- Standard Dimensions (Wrapped in IFNULL to guarantee value)
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics (Urine Analysis Data - Fallback to 'N/A' or 0)
  IFNULL(F.Urine_Colour, 'N/A') AS Urine_Colour,
  IFNULL(F.Appearance, 'N/A') AS Appearance,
  IFNULL(F.Specific_Gravity, 0.0) AS Specific_Gravity,
  IFNULL(F.pH, 0.0) AS pH,
  IFNULL(F.Proteins, 'N/A') AS Proteins,
  IFNULL(F.Glucose, 'N/A') AS Glucose,
  IFNULL(F.Bilirubin, 'N/A') AS Bilirubin,
  IFNULL(F.Ketones, 'N/A') AS Ketones,
  IFNULL(F.Blood, 'N/A') AS Blood,
  IFNULL(F.Urobilinogen, 'N/A') AS Urobilinogen,
  IFNULL(F.Nitrites, 'N/A') AS Nitrites,
  IFNULL(F.WBC_Pus_Cells, 'N/A') AS WBC_Pus_Cells,
  IFNULL(F.RBC, 'N/A') AS RBC,
  IFNULL(F.Epithelial_Cells, 'N/A') AS Epithelial_Cells,
  IFNULL(F.Casts, 'N/A') AS Casts,
  IFNULL(F.Crystals, 'N/A') AS Crystals,
  IFNULL(F.Others, 'N/A') AS Others

FROM `bigquery-tutorial-480009.A2.Fact_Urine` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.Collected_Dimension` D ON F.Collected_FK = D.ID;
```

View_Fact_CBC

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_CBC` AS
SELECT
  -- Standard Dimensions (User ID & Date for Filtering)
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Red Blood Cell Indices
  IFNULL(F.Hemoglobin, 0.0) AS Hemoglobin,
  IFNULL(F.RBC_Count, 0.0) AS RBC_Count,
  IFNULL(F.Hematocrit, 0.0) AS Hematocrit,
  IFNULL(F.MCV, 0.0) AS MCV,
  IFNULL(F.MCH, 0) AS MCH,
  IFNULL(F.MCHC, 0.0) AS MCHC,
  IFNULL(F.RDW_CV, 0.0) AS RDW_CV,
  IFNULL(F.RDW_SD, 0.0) AS RDW_SD,

  -- Metrics: White Blood Cell Counts
  IFNULL(F.WBC_Count, 0) AS WBC_Count,
  IFNULL(F.Neutrophils, 0) AS Neutrophils,
  IFNULL(F.Lymphocytes, 0) AS Lymphocytes,
  IFNULL(F.Eosinophils, 0) AS Eosinophils,
  IFNULL(F.Monocytes, 0) AS Monocytes,
  IFNULL(F.Basophils, 0.0) AS Basophils,

  -- Metrics: Absolute Counts
  IFNULL(F.Abs_Neutrophils, 0.0) AS Abs_Neutrophils,
  IFNULL(F.Abs_Lymphocytes, 0.0) AS Abs_Lymphocytes,
  IFNULL(F.Abs_Monocytes, 0.0) AS Abs_Monocytes,
  IFNULL(F.Abs_Eosinophils, 0.0) AS Abs_Eosinophils,
  IFNULL(F.Abs_Basophils, 0.0) AS Abs_Basophils

FROM `bigquery-tutorial-480009.A2.Fact_CBC` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```

View_Fact_Platet_Profile

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Platelet_Profile` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Platelet Indices
  IFNULL(F.Platelet_Count, 0) AS Platelet_Count,
  IFNULL(F.MPV, 0.0) AS MPV,
  IFNULL(F.Platelet_RDW, 0) AS Platelet_RDW,
  IFNULL(F.PCT, 0.0) AS PCT,
  IFNULL(F.P_LCR, 0.0) AS P_LCR,

  -- Metrics: Immature Platelet Fraction & Others
  IFNULL(F.IMG, 0.0) AS IMG,
  IFNULL(F.IMM, 0.0) AS IMM,
  IFNULL(F.IML, 0.0) AS IML,
  IFNULL(F.LIC, 0.0) AS LIC

FROM `bigquery-tutorial-480009.A2.Fact_Platelet_Profile` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```


View_Fact_Lipid_Profile

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Lipid_Profile` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Cholesterol Data
  IFNULL(F.Total_Cholesterol, 0) AS Total_Cholesterol,
  IFNULL(F.HDL, 0) AS HDL,
  IFNULL(F.LDL, 0.0) AS LDL,

  -- Metrics: Converted from String to Number for Graphing
  IFNULL(SAFE_CAST(F.VLDL AS FLOAT64), 0.0) AS VLDL,
  IFNULL(SAFE_CAST(F.Triglycerides AS FLOAT64), 0.0) AS Triglycerides,

  -- Metrics: Ratios and Calculations
  IFNULL(F.Non_HDL, 0) AS Non_HDL,
  IFNULL(F.Total_HDL_Ratio, 0.0) AS Total_HDL_Ratio,
  IFNULL(F.LDL_HDL_Ratio, 0.0) AS LDL_HDL_Ratio,
  IFNULL(F.HDL_LDL_Ratio, 0.0) AS HDL_LDL_Ratio

FROM `bigquery-tutorial-480009.A2.Fact_Lipid_Profile` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```

View_Fact_Liver_Function

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Liver_Function` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Bilirubin Levels
  IFNULL(F.Bilirubin_Total, 0.0) AS Bilirubin_Total,
  IFNULL(F.Bilirubin_Direct, 0.0) AS Bilirubin_Direct,
  IFNULL(F.Bilirubin_Indirect, 0.0) AS Bilirubin_Indirect,

  -- Metrics: Enzymes
  IFNULL(F.ALP, 0) AS ALP,
  IFNULL(F.ALT_SGPT, 0) AS ALT_SGPT,
  IFNULL(F.AST_SGOT, 0) AS AST_SGOT,
  IFNULL(F.GGT, 0) AS GGT,

  -- Metrics: Proteins
  IFNULL(F.Protein_Total, 0.0) AS Protein_Total,
  IFNULL(F.Albumin, 0.0) AS Albumin,
  IFNULL(F.Globulin, 0.0) AS Globulin,
  IFNULL(F.A_G_Ratio, 0.0) AS A_G_Ratio

FROM `bigquery-tutorial-480009.A2.Fact_Liver_Function` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```

View_Fact_Kidney_Function

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Kidney_Function` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Kidney Markers
  IFNULL(F.Creatinine, 0.0) AS Creatinine,
  IFNULL(F.Urea, 0.0) AS Urea,
  IFNULL(F.BUN, 0.0) AS BUN,
  IFNULL(F.BUN_Creatinine_Ratio, 0.0) AS BUN_Creatinine_Ratio,

  -- Metrics: Electrolytes (Sodium/Chloride are Ints in your schema)
  IFNULL(F.Sodium, 0) AS Sodium,
  IFNULL(F.Potassium, 0.0) AS Potassium,
  IFNULL(F.Chloride, 0) AS Chloride,

  -- Metrics: Other
  IFNULL(F.Uric_Acid, 0.0) AS Uric_Acid,
  IFNULL(F.eGFR, 0.0) AS eGFR

FROM `bigquery-tutorial-480009.A2.Fact_Kidney_Function` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```

View_Fact_Iron_Profile

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Iron_Profile` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Iron Levels (Integers)
  IFNULL(F.Iron, 0) AS Iron,
  IFNULL(F.UIBC, 0) AS UIBC,
  IFNULL(F.TIBC, 0) AS TIBC,

  -- Metrics: Saturation (Float)
  IFNULL(F.Transferrin_Saturation, 0.0) AS Transferrin_Saturation

FROM `bigquery-tutorial-480009.A2.Fact_Iron_Profile` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```

View_Fact_HbA1C

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_HbA1c` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Glycated Hemoglobin
  IFNULL(F.HbA1c, 0.0) AS HbA1c,
  IFNULL(F.Estimated_Avg_Glucose, 0.0) AS Estimated_Avg_Glucose,
  IFNULL(F.HbF, 0.0) AS HbF

FROM `bigquery-tutorial-480009.A2.Fact_HbA1c` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.Collected_Dimension` D ON F.Collected_FK = D.ID;
```

View_Fact_Urine_ACR

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Urine_ACR` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Microalbuminuria Screening
  IFNULL(F.Urine_Albumin, 0.0) AS Urine_Albumin,
  IFNULL(F.Urine_Creatinine, 0.0) AS Urine_Creatinine,
  IFNULL(F.Albumin_Creatinine_Ratio, 0.0) AS Albumin_Creatinine_Ratio

FROM `bigquery-tutorial-480009.A2.Fact_Urine_ACR` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```

View_Fact_Calcium_Phos

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Calcium_Phos` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Mineral Levels
  IFNULL(F.Calcium, 0.0) AS Calcium,
  IFNULL(F.Phosphorus, 0.0) AS Phosphorus

FROM `bigquery-tutorial-480009.A2.Fact_Calcium_Phos` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```

View_Fact_Thyroid_Profile

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Thyroid_Profile` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Thyroid Hormones
  IFNULL(F.TT3, 0) AS TT3,           -- Triiodothyronine (Int)
  IFNULL(F.TT4, 0.0) AS TT4,        -- Thyroxine (Float)
  IFNULL(F.TSH, 0.0) AS TSH         -- Thyroid Stimulating Hormone (Float)

FROM `bigquery-tutorial-480009.A2.Fact_Thyroid_Profile` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```


View_Fact_Glucose_Fasting

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Glucose_Fasting` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Blood Sugar
  IFNULL(F.Fasting_Glucose, 0) AS Fasting_Glucose

FROM `bigquery-tutorial-480009.A2.Fact_Glucose_Fasting` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```

View_Fact_Glucose_PP

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Glucose_PP` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Post-Meal Glucose
  IFNULL(F.Postprandial_Glucose, 0) AS Postprandial_Glucose

FROM `bigquery-tutorial-480009.A2.Fact_Glucose_PP` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```

View_Fact_Glucose_Diagnopath

```
CREATE OR REPLACE VIEW `bigquery-tutorial-480009.A2.View_Fact_Glucose_Diagnopath` AS
SELECT
  -- Standard Dimensions
  IFNULL(P.Original_MedID, 'Unknown') AS Original_MedID,
  IFNULL(P.First_Name, 'Unknown') AS First_Name,
  IFNULL(P.Last_Name, 'Unknown') AS Last_Name,
  IFNULL(P.Gender, 'Unknown') AS Gender,
  IFNULL(P.Age_Category, 'Unknown') AS Age_Category,
  IFNULL(D.Date, DATE('1970-01-01')) AS Test_Date,

  -- Metrics: Diagnopath Glucose Tests
  IFNULL(F.FBS, 0) AS FBS,    -- Fasting Blood Sugar
  IFNULL(F.PLBS, 0) AS PLBS  -- Post-Lunch Blood Sugar

FROM `bigquery-tutorial-480009.A2.Fact_Glucose_Diagnopath` F
JOIN `bigquery-tutorial-480009.A2.MedID_Dimension` P ON F.MedID_FK = P.ID
JOIN `bigquery-tutorial-480009.A2.CollectedException` D ON F.CollectedException_FK = D.ID;
```

Schema

MedID_Dimension

Column Name	Data Type	Nullable
ID	INT64	NO
Original_MedID	STRING	NO
LabReference	STRING	NO
First_Name	STRING	NO
Last_Name	STRING	NO
Age	INT64	NO
Age_Category	STRING	NO
Gender	STRING	NO
Diet	STRING	NO
Registration_Weight	FLOAT64	NO

LabReference_Dimension

Column Name	Data Type	Nullable
ID	INT64	NO
LabReference	STRING	NO

SampleID_Dimension

Column Name	Data Type	Nullable
ID	INT64	NO
Sample_ID	STRING	NO

Collected_Dimension

Column Name	Data Type	Nullable
ID	INT64	NO
Date	DATE	NO

Time_Dimension

Column Name	Data Type	Nullable
ID	INT64	NO
Time	TIME	NO

Reported_Time_Dimension

Column Name	Data Type	Nullable
ID	INT64	NO
Reported_Time	TIME	NO

Fact_Urine

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- Urine_Colour - STRING
- Appearance - STRING
- Specific_Gravity - FLOAT64
- pH - FLOAT64
- Proteins - STRING
- Glucose - STRING
- Bilirubin - STRING
- Ketones - STRING
- Blood - STRING
- Urobilinogen - STRING
- Nitrites - STRING
- WBC_Pus_Cells - STRING
- RBC - STRING
- Epithelial_Cells - STRING
- Casts - STRING
- Crystals - STRING
- Others - STRING

Fact_CBC

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- Hemoglobin - FLOAT64
- RBC_Count - FLOAT64
- Hematocrit - FLOAT64

- MCV - FLOAT64
- MCH - INT64
- MCHC - FLOAT64
- RDW_CV - FLOAT64
- RDW_SD - FLOAT64
- WBC_Count - INT64
- Neutrophils - INT64
- Lymphocytes - INT64
- Eosinophils - INT64
- Monocytes - INT64
- Basophils - FLOAT64
- Abs_Neutrophils - FLOAT64
- Abs_Lymphocytes - FLOAT64
- Abs_Monocytes - FLOAT64
- Abs_Eosinophils - FLOAT64
- Abs_Basophils - FLOAT64

Fact_Platelet_Profile

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- Platelet_Count - INT64
- MPV - FLOAT64
- Platelet_RDW - INT64
- PCT - FLOAT64
- P_LCR - FLOAT64
- IMG - FLOAT64
- IMM - FLOAT64
- IML - FLOAT64
- LIC - FLOAT64

Fact_Lipid_Profile

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- Total_Cholesterol - INT64

- HDL - INT64
- LDL - FLOAT64
- VLDL - STRING
- Triglycerides - STRING
- Non_HDL - INT64
- Total_HDL_Ratio - FLOAT64
- LDL_HDL_Ratio - FLOAT64
- HDL_LDL_Ratio - FLOAT64

Fact_Liver_Function

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- Bilirubin_Total - FLOAT64
- Bilirubin_Direct - FLOAT64
- Bilirubin_Indirect - FLOAT64
- ALP - INT64
- ALT_SGPT - INT64
- AST_SGOT - INT64
- GGT - INT64
- Protein_Total - FLOAT64
- Albumin - FLOAT64
- Globulin - FLOAT64
- A_G_Ratio - FLOAT64

Fact_Kidney_Function

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- Creatinine - FLOAT64
- Urea - FLOAT64
- BUN - FLOAT64
- BUN_Creatinine_Ratio - FLOAT64
- Sodium - INT64
- Potassium - FLOAT64
- Chloride - INT64

- Uric_Acid - FLOAT64
- eGFR - FLOAT64

Fact_Iron_Profile

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- Iron - INT64
- UIBC - INT64
- TIBC - INT64
- Transferrin_Saturation - FLOAT64

Fact_HbA1c

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- HbA1c - FLOAT64
- Estimated_Avg_Glucose - FLOAT64
- HbF - FLOAT64

Fact_Urine_ACR

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- Urine_Albumin - FLOAT64
- Urine_Creatinine - FLOAT64
- Albumin_Creatinine_Ratio - FLOAT64

Fact_Calcium_Phos

- MedID_FK - INT64
- LabReference_FK - INT64

- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- Calcium - FLOAT64
- Phosphorus - FLOAT64

Fact_Thyroid_Profile

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- TT3 - INT64
- TT4 - FLOAT64
- TSH - FLOAT64

Fact_Glucose_Fasting

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- Fasting_Glucose - INT64

Fact_Glucose_PP

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64
- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- Postprandial_Glucose - INT64

Fact_Glucose_Diagnopath

- MedID_FK - INT64
- LabReference_FK - INT64
- SampleID_FK - INT64

- Collected_FK - INT64
- Time_FK - INT64
- Reported_Time_FK - INT64
- FBS - INT64
- PLBS - INT64